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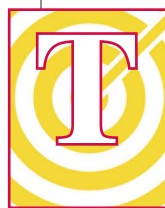


Clarify Your Indicators Using Kendall Correlation

Noise Elimination Technology

Reduce indicator noise and clarify the direction of your indicators using Kendall rank correlation, which measures the strength of dependence between two sets of variables. Here's how you can use it.

by John F. Ehlers



The purpose of technical indicators is to provide an insight into the inefficiencies in the market to make better-informed trading decisions. The problem with most indicators is they use only a short amount of data and are, therefore, noisy themselves. Noise can be removed by smoothing filters, but smoothing

filters can cause indicator lag and delayed decisions. These delays can be costly.

I will show you how to clarify your indicators without filtering. This is accomplished using the nonlinearity of a rank-ordered Kendall correlation.

KENDALL CORRELATION

In general, *Kendall correlation* compares the ranked order of two sets of random variables. Pairs of ranked variables can be either concordant or discordant. I'll leave the details of the Kendall correlation description to the statistics tutorials or Wikipedia. The correlation equation is:

$$\tau = \frac{2}{n(n-1)} \sum_{i < j} \text{sgn}(x_i - x_j) \text{sgn}(y_i - y_j)$$

When used as noise elimination technology, the y variable is a straight line with a positive slope and the x variable is the indicator output. The Kendall correlation in this configuration basically strips out the noise components not going in the main direction of the indicator in a nonlinear fashion.

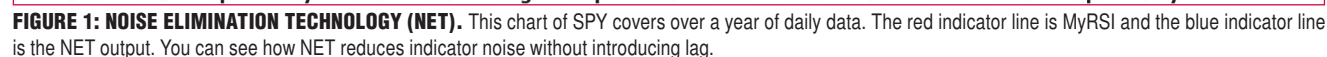
NOISE ELIMINATION TECHNOLOGY EXAMPLE

The best way to describe NET is with an example. I will start with MyRSI, a version of the RSI indicator that I derived in my article in the May 2018 issue of STOCKS & COMMODITIES. I prefer this version because it provides for the RSI to swing between -1 and +1.

This formulation is easier to use when dealing with market cycles. The computation accrues *closes up* (CU) and *closes down* (CD). The MyRSI indicator is simply the ratio of their difference divided by their sum. This is a relatively noisy indicator.

In the sidebar "EasyLanguage code for noise elimination technology," I provide coding in EasyLanguage for implementing NET using the MyRSI indicator for the example.

The second block of code given in the sidebar is the noise



simplified from the general equation because $\text{sgn}(y_i - y_j)$ is always negative since $Y[]$ is a straight line. The denominator is a simple calculation, and NET is the ratio of the numerator to the denominator.

The effectiveness of the nonlinear noise elimination technology is demonstrated in Figure 1. The candlestick chart is over a year's worth of daily data on the SPDR S&P 500 ETF (SPY). I have used the standard 14-day length to compute MyRSI,

```
Plot2(NET);
```


and the same length for NET. The red indicator line is MyRSI, and the blue indicator line is the NET output. Clearly, the goal of reducing the indicator noise without introducing lag has been obtained.

NET does not necessarily replace smoothing filters, but it can work together with them to provide the best visual presentation. I hope that applying NET to your favorite indicators will help improve the clarity of your trading.

John Ehlers, a Contributing Editor to STOCKS & COMMODITIES, is a pioneer in the use of cycles and DSP (digital signal processing) technical analysis. He is president of MESA Software. He can be reached through his website at MESA-software.com.

*The code given in this article is available in the **Article Code** section of our website, Traders.com.*

*See our **Traders' Tips** section beginning on page 48 for implementation of John Ehlers' technique in various technical analysis programs. Accompanying program code can be found in the Traders' Tips section at Traders.com.*

The Kendall correlation basically strips out the noise components not going in the main direction of the indicator.



FURTHER READING

Ehlers, John F. [2013]. *Cycle Analytics For Traders*, John Wiley & Sons.

—— [2018]. “RocketRSI—A Solid Propellant For Your Rocket Science Trading,” *Technical Analysis of STOCKS & COMMODITIES*, Volume 36: May.

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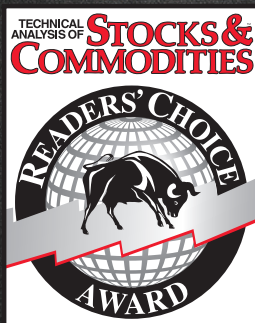
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